

DS 5110 Big Data Systems

Lab: AWS Services I

Topic: Basics of AWS Identity and Access Management (IAM)

Last Updated: February 25, 2026

Learning Objectives:

- Demonstrate the creation of an S3 bucket and save a file
- Understand how to create an IAM role
- Understand how to create an IAM policy
- Understand how to spin up an EC2 instance
- Explain the purpose of an IAM role
- Explain the purpose of IAM policies
- Explain the purpose of an Amazon Resource Name (ARN)

Submission: Save all of your results and screenshots in a file (Word or PDF doc). Number the steps, keeping things clear and organized.

Total Points: 9

Link to AWS Management Console:

<https://aws.amazon.com/console/>

Assigned Reading

Policies and permissions in IAM (up to but excluding IAM permissions boundaries)

https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies.html

Roles

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Policies

A policy is an object in AWS that defines the permissions of an identity or resource.

Instructions

This exercise will have you working with various AWS services.

Specifically, you will:

1. Visit S3, create a bucket, and upload a file to it.
2. Create an IAM role and assign policy that allows you to write to and read from the bucket.
3. Spin up an EC2 instance using the role, and write a CLI command to show the contents of the file. Along the way, you will take screenshots of output to submit.

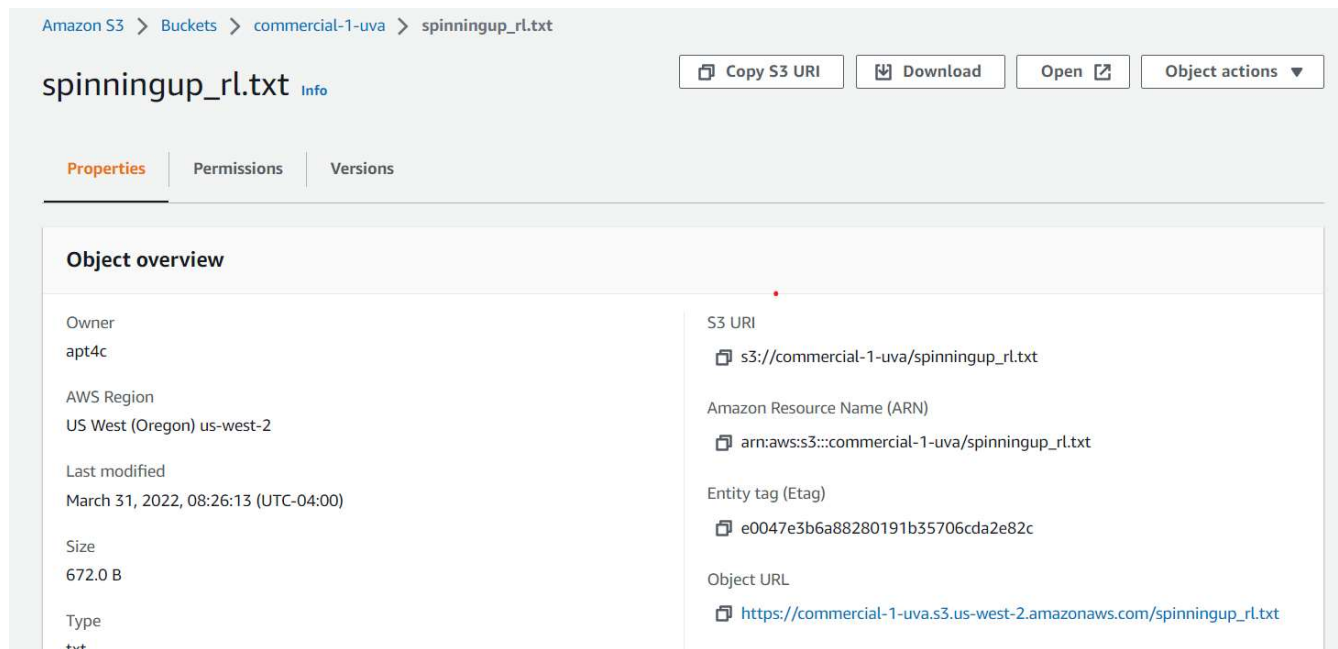
Follow the steps very carefully.

1. **(1 POINT)** Go to the **S3** service. Create a new bucket and upload the file: *spinningup_rl.txt*

FYI: This is a bootstrap file that installs python, some modules, and a repo.

Take a screenshot showing that the file is in the bucket.

2. **(1 POINT)** Take note of the S3 URI which is the path to the file on S3. You will use this in EC2 later to read from the file. Capture this URI in your results file.



The screenshot shows the Amazon S3 console interface for the object `spinningup_rl.txt` in the bucket `commercial-1-uva`. The breadcrumb navigation at the top reads: `Amazon S3 > Buckets > commercial-1-uva > spinningup_rl.txt`. The object name `spinningup_rl.txt` is displayed with an `Info` link. Action buttons include `Copy S3 URI`, `Download`, `Open`, and `Object actions`. Below the navigation tabs (`Properties`, `Permissions`, `Versions`), the `Object overview` section is expanded, showing the following details:

Property	Value
Owner	apt4c
AWS Region	US West (Oregon) us-west-2
Last modified	March 31, 2022, 08:26:13 (UTC-04:00)
Size	672.0 B
Type	txt
S3 URI	s3://commercial-1-uva/spinningup_rl.txt
Amazon Resource Name (ARN)	arn:aws:s3:::commercial-1-uva/spinningup_rl.txt
Entity tag (Etag)	e0047e3b6a88280191b35706cda2e82c
Object URL	https://commercial-1-uva.s3.us-west-2.amazonaws.com/spinningup_rl.txt

3. Go to the **IAM** service. Create a role called *data_scientist_s3_role** where * can be other characters

4. Allow common use case: EC2

Select trusted entity [Info](#)

Trusted entity type

☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.

☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.

☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.

☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.

☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Common use cases

- ☐ **EC2**
Allows EC2 instances to call AWS services on your behalf.
- ☐ **Lambda**
Allows Lambda functions to call AWS services on your behalf.

5. Create a policy that can read from/write to an S3 bucket. Specifically it should have this access:

- **GetObject** (from Read). This grants permission to retrieve objects from S3.
- **PutObject** (from Write). This grants permission to add an object to a bucket.

Create policy

1 2 3

A policy defines the AWS permissions that you can assign to a user, group, or role. You can create and edit a policy in the visual editor and using JSON. [Learn more](#)

Visual editor **JSON** [Import managed policy](#)

Expand all | Collapse all

▼ Select a service [Clone](#) [Remove](#)

▼ Service [close](#)

Select a service below

[Access Analyzer ?](#) [Elastic Transcoder ?](#) [Mobile Hub ?](#)

[Enter service manually](#)

You will want to provide permissions to a specific file. To do this, specify an Amazon Resource Name (ARN) for a specific S3 bucket and object. You will need to have at least one bucket with a file created in S3; please create one if needed. When I enter bucket name and object name, the ARN is auto-populated.

Add ARN(s)

Amazon Resource Names (ARNs) uniquely identify AWS resources. Resources are unique to each service. [Learn more](#)

Specify ARN for object [List ARNs manually](#)

arn:aws:s3:::commercial-1-uva/spinningup_rl.txt

Bucket name *

commercial-1-uva
☐ Any

Object name *

spinningup_rl.txt
☐ Any

Cancel
Add

6. **(1 POINT)** Take a screenshot of the policy summary page that shows the created policy. You'll need to give the policy and name. Take note so you can find it for the next step. You can see mine is called *data_scientist_s3_policy*

Policies > data_scientist_s3_policy
Delete policy

Summary

Policy ARNarn:aws:iam::208637696492:policy/data_scientist_s3_policy
Description

Permissions
Policy usage
Tags
Policy versions
Access Advisor

Policy summary
{} JSON
Edit policy
?

Filter

Service	Access level	Resource	Request condition
Allow (1 of 371 services) Show remaining 370			
S3	Limited: Read, Write	ObjectPath string like spinningup_rl.txt, BucketName string like commercial-1-uva	None

7. Attach the policy to the role.

IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Add permissions [Info](#)

Permissions policies (Selected 1/826) [Info](#)

Choose one or more policies to attach to your new role.

Filter policies by property or policy name and press enter.

< 1 2 3 4 5 6 7 ... 42 > [Settings](#)

☐	Policy name ↗	Type ▼	Description
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☒	+ data_scientist_s3_policy	Custom...	
-------------------------------------	--	-----------	--

8. The final role creation step should look like this:

Role name

Enter a meaningful name to identify this role.

data_scientist_s3_role

Maximum 64 characters. Use alphanumeric and '+', '@', '-' characters.

Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use alphanumeric and '+', '@', '-' characters.

Step 1: Select trusted entities

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "sts:AssumeRole"  
8       ],  
9       "Principal": {  
10        "Service": [  
11          "ec2.amazonaws.com"  
12        ]  
13      }  
14    }  
15  ]  
16 }
```

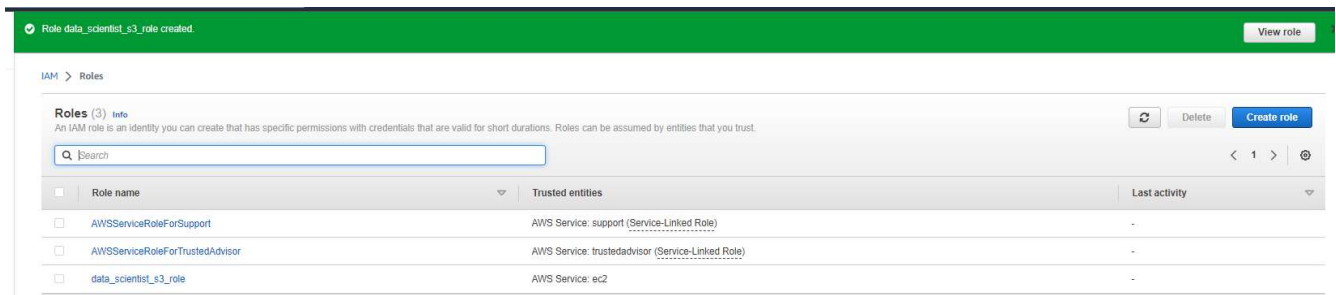
Step 2: Add permissions

Permissions policy summary

Policy name ↗	Type ▼	Attached as
data_scientist_s3_policy	Customer managed	Permissions policy

Notice the trusted entities (EC2) is pre-populated in JSON.

9. (1 POINT) Take a screenshot of the Roles page to show the new role.



10. (1 POINT) Create an EC2 instance with these parameters:

-size: t3.micro (this is free on the AWS Free Tier)

-OS: Amazon Linux

-create or select a key pair

-under Network settings, select Allow HTTPS traffic from the internet. (THIS WILL ENABLE INSTANCE CONNECT)

▼ **Network settings** [Info](#) Edit

Network [Info](#)

vpc-f19af88c

Subnet [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-31' with the following rules:

☒ Allow SSH traffic from Helps you connect to your instance Anywhere
0.0.0.0/0

☒ Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server

-under Advanced details > IAM instance profile, select the role you've created (THIS IS IMPORTANT)

▼ **Advanced details** [Info](#)

Purchasing option [Info](#)

☐ Request Spot Instances
Request Spot Instances at the Spot price, capped at the On-Demand price

Domain join directory [Info](#)

Select ▼ [Create new directory](#)

IAM instance profile [Info](#)

data_scientist_s3_role
arn:aws:iam::208637696492:instance-profile/data_scientist_s3_role ▼ [Create new IAM profile](#)

Take a screenshot showing the instance.

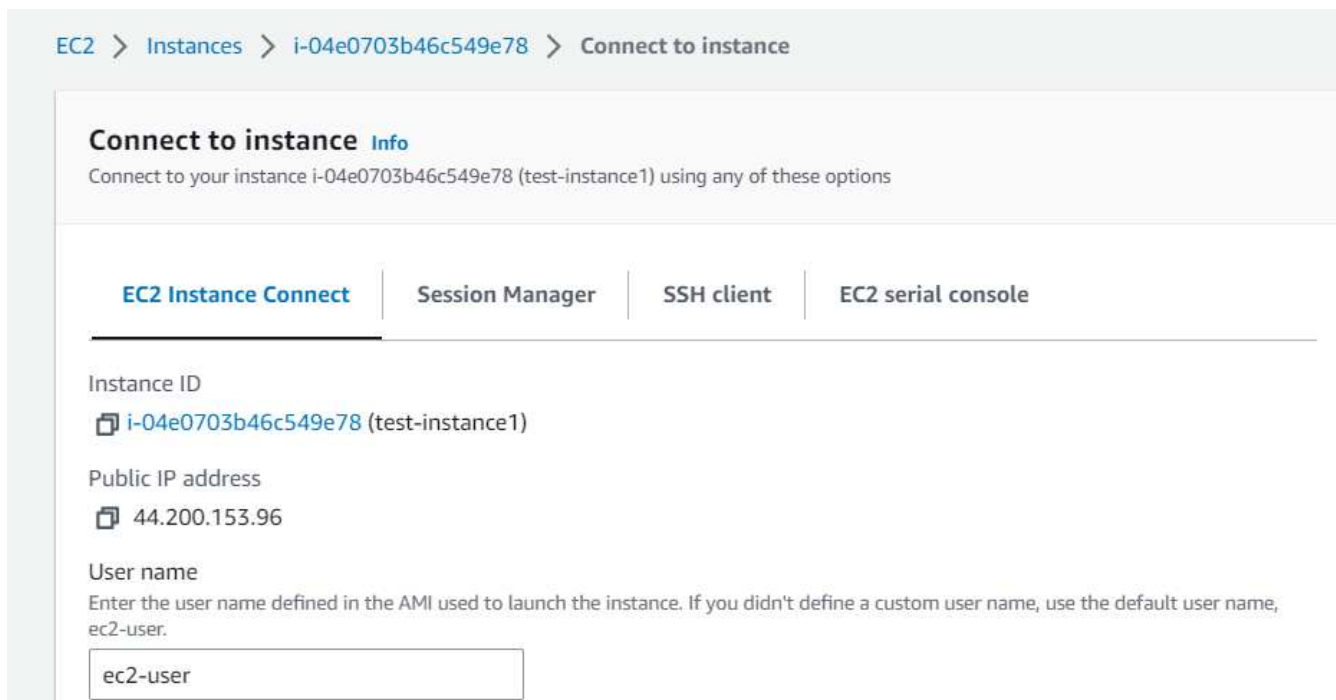
11. Launch the instance

12. Connect to the instance.

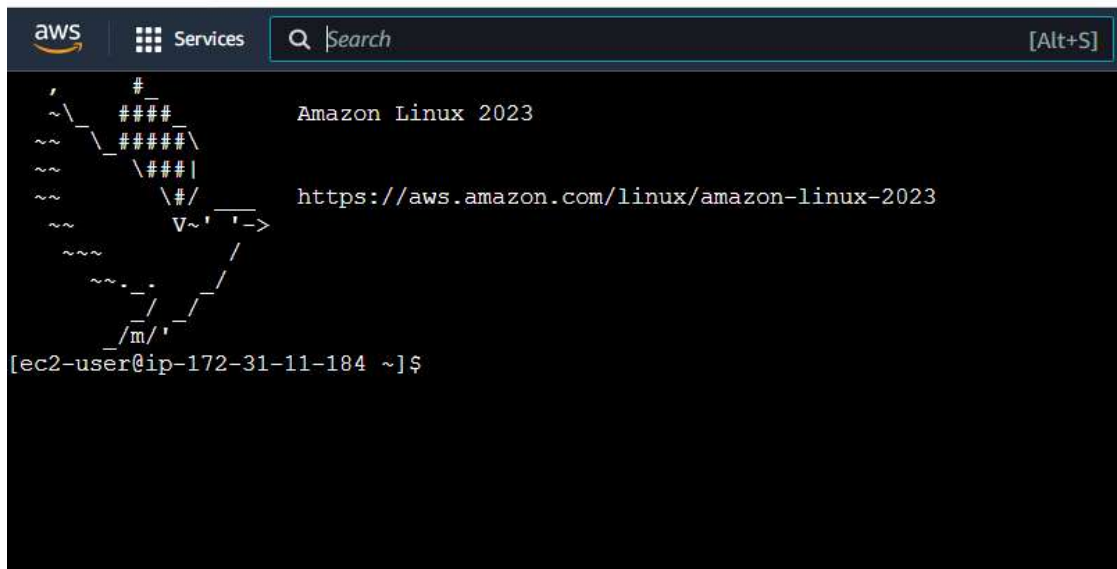
There are several options for connection including:

-EC2 Instance Connect (through the AWS Management Console)

-SSH client (on a Mac you can use Terminal; on a Windows machine you can use PuTTY)



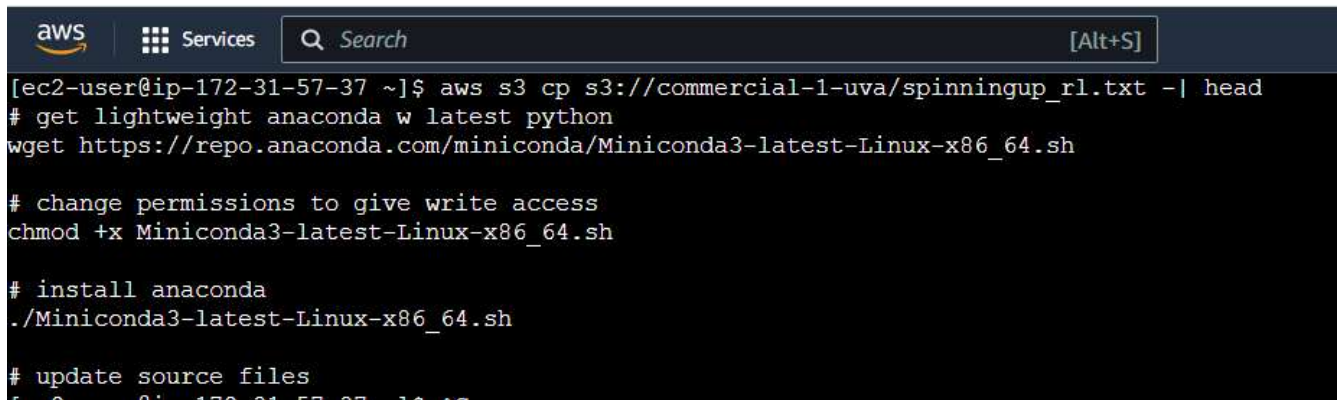
13. (1 POINT) You will see the landing page of your EC2 instance. Take a screenshot.



14. (1 POINT) Run the command following prompt \$ at the terminal to show the top lines of the file you've uploaded to S3. This uses the aws s3 CLI tool to copy the file contents and pipe to a file that shows the head. The portion `s3://commercial-1-uva/spinningup_rl.txt` is my S3 URI.

```
[ec2-user@ip-172-31-57-37 ~]$ aws s3 cp s3://commercial-1-uva/spinningup_rl.txt - | head
```

Take a screenshot of the command and the output, which should look like this:



```
aws Services Search [Alt+S]
[ec2-user@ip-172-31-57-37 ~]$ aws s3 cp s3://commercial-1-uva/spinningup_r1.txt -| head
# get lightweight anaconda w latest python
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

# change permissions to give write access
chmod +x Miniconda3-latest-Linux-x86_64.sh

# install anaconda
./Miniconda3-latest-Linux-x86_64.sh

# update source files
```

15. Run the top lines of code at the command line to install Anaconda. You might copy/paste. They are reproduced here. You'll need to agree and press enter a few times to complete installation. When it completes, you'll see the screenshot below.

```
# get lightweight anaconda w latest python
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

# change permissions to give write access
chmod +x Miniconda3-latest-Linux-x86_64.sh

# install anaconda
./Miniconda3-latest-Linux-x86_64.sh
```

```
aws Services Search [Alt+S]

Downloading and Extracting Packages

Downloading and Extracting Packages

Preparing transaction: done
Executing transaction: done
installation finished.
Do you wish the installer to initialize Miniconda3
by running conda init? [yes|no]
[no] >>> yes
no change      /home/ec2-user/miniconda3/condabin/conda
no change      /home/ec2-user/miniconda3/bin/conda
no change      /home/ec2-user/miniconda3/bin/conda-env
no change      /home/ec2-user/miniconda3/bin/activate
no change      /home/ec2-user/miniconda3/bin/deactivate
no change      /home/ec2-user/miniconda3/etc/profile.d/conda.sh
no change      /home/ec2-user/miniconda3/etc/fish/conf.d/conda.fish
no change      /home/ec2-user/miniconda3/shell/condabin/Conda.ps1
no change      /home/ec2-user/miniconda3/shell/condabin/conda-hook.ps1
no change      /home/ec2-user/miniconda3/lib/python3.10/site-packages/xontrib/conda.xsh
no change      /home/ec2-user/miniconda3/etc/profile.d/conda.csh
modified       /home/ec2-user/.bashrc

==> For changes to take effect, close and re-open your current shell. <==

If you'd prefer that conda's base environment not be activated on startup,
set the auto_activate_base parameter to false:

conda config --set auto_activate_base false

Thank you for installing Miniconda3!
[ec2-user@ip-172-31-57-37 ~]$
```

16. Run this line to update your session:

```
source ~/.bashrc
```

17. **(1 POINT)** Next, you'll show that python is installed by running at the command line:

```
$ python
```

This should launch the latest Python. It's now available for the lifetime of this instance. Take a screenshot.

```
(base) [ec2-user@ip-172-31-57-37 ~]$ python
Python 3.10.9 (main, Jan 11 2023, 15:21:40) [GCC 11.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> 2 + 4
6
>>>
```

18. **(1 POINT)** We are finished with this session and instance. Revisit the instances and terminate your EC2 instance. Take a screenshot that shows your instance has been terminated.

Successfully terminated i-063c675f46a1489ca

Instances (2) Info

Find instance by attribute or tag (case-sensitive)

Refresh

Connect

Instance state

Actions

Launch instances

<

1

>

Settings

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4
☐	test-instance1	i-063c675f46a1489ca	Terminated	t2.micro	-	No alarms	us-east-1e	-	-

If you were able to complete this assignment, I commend you! There's a lot to learn when getting started with AWS.